```
import subprocess
import sys
lib = ["numpy", "pandas", "matplotlib"]
for l in lib:
    subprocess.check_call([sys.executable, "-m", "pip","install", l])
print("Success")
```

```
Success
```

```
try:
    import numpy as np
    import pandas as pd
    print("imported")
except ImportError as e:
    print("Import Error : ", e)
```

```
imported
```

```
import pandas as pd
try:
    df = pd.read_csv(r'data.csv')
    print(df)
except FileNotFoundError as e:
    print("Not Found:",e)
```

```
          Name  age
0      shankar   12
1          ram   14
2        arjun   18
3         luan   19
4      shankar   20
5      srivani   21
6  mallikarjun   40
7        meena   40
```

```
#week-2
import pandas as pd
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Emma'],
    'Age': [20, 22, 21, 23, 20],
    'Marks': [85.5, 72.0, 90.0, 65.5, 88.0],
    'Department': ['CSE', 'ECE', 'CSE', 'EEE', 'ECE']
}
df = pd.DataFrame(data)
print(df)
print(df.head())

print("Data Types :")
print(df.dtypes)

print("Dataset Info: ")
```

```python
print(df.info())

print(df.dtypes)
df["Age"] = df["Age"].astype(str)
print(df.dtypes)
```

```
      Name  Age  Marks Department
0    Alice   20   85.5        CSE
1      Bob   22   72.0        ECE
2  Charlie   21   90.0        CSE
3    David   23   65.5        EEE
4     Emma   20   88.0        ECE
      Name  Age  Marks Department
0    Alice   20   85.5        CSE
1      Bob   22   72.0        ECE
2  Charlie   21   90.0        CSE
3    David   23   65.5        EEE
4     Emma   20   88.0        ECE
Data Types :
Name              str
Age             int64
Marks         float64
Department        str
dtype: object
Dataset Info:
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 4 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Name        5 non-null      str
 1   Age         5 non-null      int64
 2   Marks       5 non-null      float64
 3   Department  5 non-null      str
dtypes: float64(1), int64(1), str(2)
memory usage: 292.0 bytes
None
Name              str
Age             int64
Marks         float64
Department        str
dtype: object
Name              str
Age               str
Marks         float64
Department        str
dtype: object
```

```python
#week-3
# Create a dataset with missing values, duplicates, and an outlier
import pandas as pd
import numpy as np
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David',
             'Alice', 'Emma', 'Frank'],
    'Age': [20, 22, np.nan, 23, 20, 21, 120],
    'Marks': [85, 72, 90, np.nan, 85, 88, 65]
}
print("Original DataSet : ")
df = pd.DataFrame(data)
print(df)
print("Missing Values : ")
print(df.isnull().sum())
print("Fill the null values :")
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())
print(df)
print("Duplicates rows:")
print(df.duplicated().sum())
df = df.drop_duplicates()
print("Remove duplicate row :")
print(df)
print("OutLiers : ")
q1 = df["Age"].quantile(0.25)
q3 = df["Age"].quantile(0.75)
iqr = q3 - q1
low = q1 - 1.5*iqr
up = q3 + 1.5*iqr
outliers = df[(df["Age"] < low) | (df["Age"]>up)]
print(outliers)

median_age = df['Age'].median()

df.loc[(df['Age'] < low) | (df['Age'] > up),'Age'] = median_age

print("Dataset after handling outliers:")
print(df)
print("Final Cleaned Dataset:")
print(df)

print("\nMissing values remaining:")
print(df.isnull().sum())

print("\nDuplicate rows remaining:")
print(df.duplicated().sum())
```

```
Original DataSet :
      Name    Age  Marks
```

```
0    Alice   20.0   85.0
1      Bob   22.0   72.0
2  Charlie    NaN   90.0
3    David   23.0    NaN
4    Alice   20.0   85.0
5     Emma   21.0   88.0
6    Frank  120.0   65.0
Missing Values :
Name     0
Age      1
Marks    1
dtype: int64
Fill the null values :
      Name         Age      Marks
0    Alice   20.000000  85.000000
1      Bob   22.000000  72.000000
2  Charlie   37.666667  90.000000
3    David   23.000000  80.833333
4    Alice   20.000000  85.000000
5     Emma   21.000000  88.000000
6    Frank  120.000000  65.000000
Duplicates rows:
1
Remove duplicate row :
      Name         Age      Marks
0    Alice   20.000000  85.000000
1      Bob   22.000000  72.000000
2  Charlie   37.666667  90.000000
3    David   23.000000  80.833333
5     Emma   21.000000  88.000000
6    Frank  120.000000  65.000000
OutLiers :
    Name    Age  Marks
6  Frank  120.0   65.0
Dataset after handling outliers:
      Name        Age      Marks
0    Alice  20.000000  85.000000
1      Bob  22.000000  72.000000
2  Charlie  37.666667  90.000000
3    David  23.000000  80.833333
5     Emma  21.000000  88.000000
6    Frank  22.500000  65.000000
Final Cleaned Dataset:
      Name        Age      Marks
0    Alice  20.000000  85.000000
1      Bob  22.000000  72.000000
2  Charlie  37.666667  90.000000
3    David  23.000000  80.833333
5     Emma  21.000000  88.000000
```

```
6    Frank  22.500000  65.000000

Missing values remaining:
Name     0
Age      0
Marks    0
dtype: int64

Duplicate rows remaining:
0
```

```python
#week - 4

import pandas as pd

# Create a sample dataset
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Emma'],
    'Age': [20, 22, 21, 23, 20],
    'Marks': [85, 72, 90, 65, 88],
    'Department': ['CSE', 'ECE', 'CSE', 'EEE', 'ECE']
}

df = pd.DataFrame(data)

print("Original Dataset:")
print(df)

print("Descriptive Statistics:")
print(df.describe())

print("Mean Marks:", df['Marks'].mean())
print("Median Marks:", df['Marks'].median())
print("Mode Marks:", df['Marks'].mode().tolist())
print("Minimum Marks:", df['Marks'].min())
print("Maximum Marks:", df['Marks'].max())
print("Standard Deviation:", df['Marks'].std())

print("Department Frequency:")
print(df['Department'].value_counts())

print("\nUnique Departments:")
print(df['Department'].unique())

print("\nNumber of Unique Departments:")
print(df['Department'].nunique())

print("Dataset Information:")
df.info()

print("\nMissing Values:")
```

```python
print(df.isnull().sum())

print("\nStatistical Summary:")
print(df.describe())

print("\nDepartment-wise Average Marks:")
print(df.groupby('Department')['Marks'].mean())
```

```
Original Dataset:
      Name  Age  Marks Department
0    Alice   20     85        CSE
1      Bob   22     72        ECE
2  Charlie   21     90        CSE
3    David   23     65        EEE
4     Emma   20     88        ECE
Descriptive Statistics:
             Age      Marks
count   5.00000   5.000000
mean   21.20000  80.000000
std     1.30384  10.931606
min    20.00000  65.000000
25%    20.00000  72.000000
50%    21.00000  85.000000
75%    22.00000  88.000000
max    23.00000  90.000000
Mean Marks: 80.0
Median Marks: 85.0
Mode Marks: [65, 72, 85, 88, 90]
Minimum Marks: 65
Maximum Marks: 90
Standard Deviation: 10.931605554537724
Department Frequency:
Department
CSE    2
ECE    2
EEE    1
Name: count, dtype: int64

Unique Departments:
<StringArray>
['CSE', 'ECE', 'EEE']
Length: 3, dtype: str

Number of Unique Departments:
3
Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 4 columns):
 #   Column      Non-Null Count  Dtype
```

```
---  ------      --------------  -----
 0   Name        5 non-null      str
 1   Age         5 non-null      int64
 2   Marks       5 non-null      int64
 3   Department  5 non-null      str
dtypes: int64(2), str(2)
memory usage: 292.0 bytes

Missing Values:
Name          0
Age           0
Marks         0
Department    0
dtype: int64

Statistical Summary:
             Age      Marks
count   5.00000   5.000000
mean   21.20000  80.000000
std     1.30384  10.931606
min    20.00000  65.000000
25%    20.00000  72.000000
50%    21.00000  85.000000
75%    22.00000  88.000000
max    23.00000  90.000000

Department-wise Average Marks:
Department
CSE    87.5
ECE    80.0
EEE    65.0
Name: Marks, dtype: float64
```

```python
#week - 5

import pandas as pd
import numpy as np

# Create a sample dataset
data = {
    'Study_Hours': [1, 2, 3, 4, 5],
    'Marks': [40, 50, 60, 70, 80],
    'Sleep_Hours': [8, 7, 6, 5, 4]
}

df = pd.DataFrame(data)

print("Dataset:")
print(df)
```

```
correlation = df.corr()

print("Correlation Matrix:")
print(correlation)

covariance = df.cov()

print("Covariance Matrix:")
print(covariance)

corr_value = df['Study_Hours'].corr(df['Marks'])
cov_value = df['Study_Hours'].cov(df['Marks'])

print("Correlation between Study Hours and Marks:", corr_value)
print("Covariance between Study Hours and Marks:", cov_value)

corr_matrix = np.corrcoef(
    df['Study_Hours'],
    df['Marks']
)

print("Correlation using NumPy:")
print(corr_matrix)
```

```
Dataset:
   Study_Hours  Marks  Sleep_Hours
0            1     40            8
1            2     50            7
2            3     60            6
3            4     70            5
4            5     80            4
Correlation Matrix:
             Study_Hours  Marks  Sleep_Hours
Study_Hours          1.0    1.0         -1.0
Marks                1.0    1.0         -1.0
Sleep_Hours         -1.0   -1.0          1.0
Covariance Matrix:
             Study_Hours  Marks  Sleep_Hours
Study_Hours          2.5   25.0         -2.5
Marks               25.0  250.0        -25.0
Sleep_Hours         -2.5  -25.0          2.5
Correlation between Study Hours and Marks: 1.0
Covariance between Study Hours and Marks: 25.0
Correlation using NumPy:
[[1. 1.]
 [1. 1.]]
```

```
#week - 6


import pandas as pd
```

```
import matplotlib.pyplot as plt
import seaborn as sns

# Load the employee dataset
df = pd.read_csv("employee.csv")

# 1. Display dataset
print("First five records:")
print(df.head())

print("\nDataset shape:")
print(df.shape)

print("\nData types:")
print(df.dtypes)

# 2. Descriptive statistics
print("\nStatistical Summary:")
print(df.describe())

# 3. Check missing values and duplicates
print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:", df.duplicated().sum())

# 4. Department distribution
print("\nDepartment Count:")
print(df['Department'].value_counts())

# 5. Histogram of employee ages
plt.figure(figsize=(7, 4))
sns.histplot(data=df, x='Age', bins=6, kde=True)
plt.title("Distribution of Employee Age")
plt.show()

# 6. Salary distribution by department
plt.figure(figsize=(9, 4))
sns.barplot(data=df, x='Department', y='Salary')
plt.title("Salary by Department")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 7. Experience versus salary
plt.figure(figsize=(7, 4))
sns.scatterplot(data=df, x='Experience', y='Salary')
plt.title("Experience vs Salary")
plt.show()
```

```python
# 8. Correlation heatmap
plt.figure(figsize=(7, 5))
corr = df[['Age', 'Salary', 'Experience']].corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f')
plt.title("Employee Data Correlation")
plt.show()

# 9. Department-wise average salary
print("\nAverage Salary by Department:")
print(df.groupby('Department')['Salary'].mean())
```

```
First five records:
   Emp_ID   Name Department  Age  Salary  Experience
0       1   KING        CSE   10  100000           2
1       2   MAHI         IT   20  200000           3
2       3  DHONI        CSM   30  300000           4
3       4  CHIRU         DS   25  400000           5
4       5  ROHIT        CHE   40  500000           6

Dataset shape:
(9, 6)

Data types:
Emp_ID        int64
Name            str
Department      str
Age           int64
Salary        int64
Experience    int64
dtype: object

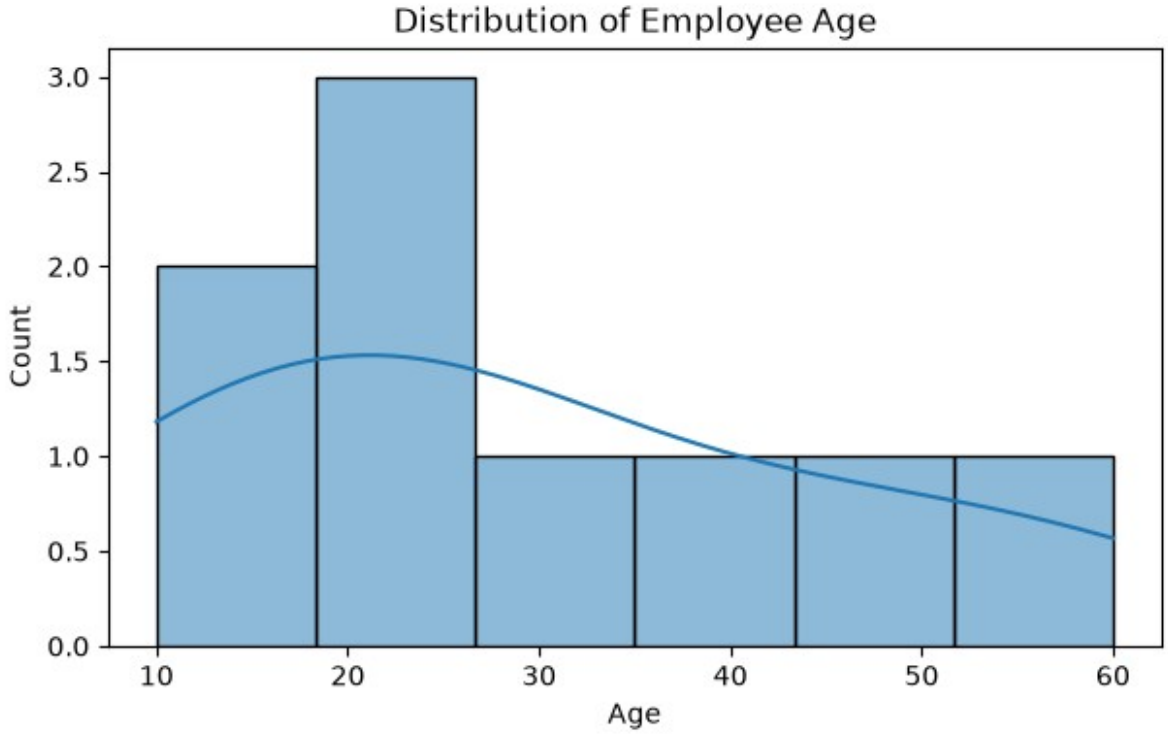
Statistical Summary:
         Emp_ID        Age         Salary  Experience
count  9.000000   9.000000       9.000000    9.000000
mean   5.000000  29.444444  344444.444444    3.666667
std    2.738613  17.400511  181046.341520    2.000000
min    1.000000  10.000000  100000.000000    1.000000
25%    3.000000  20.000000  200000.000000    2.000000
50%    5.000000  25.000000  400000.000000    3.000000
75%    7.000000  40.000000  500000.000000    5.000000
max    9.000000  60.000000  600000.000000    7.000000

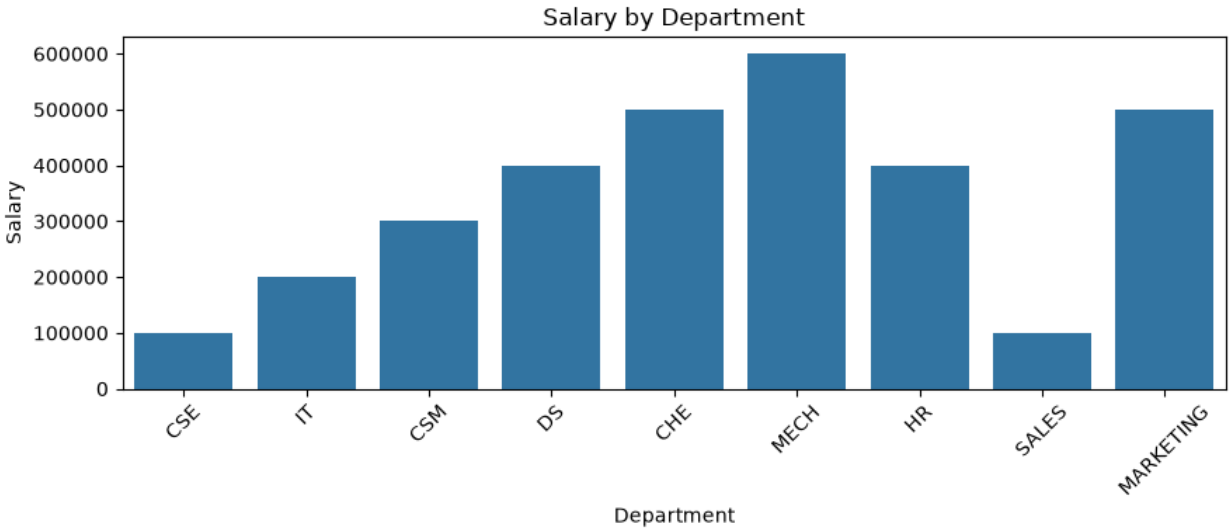
Missing Values:
Emp_ID        0
Name          0
Department    0
Age           0
Salary        0
Experience    0
```

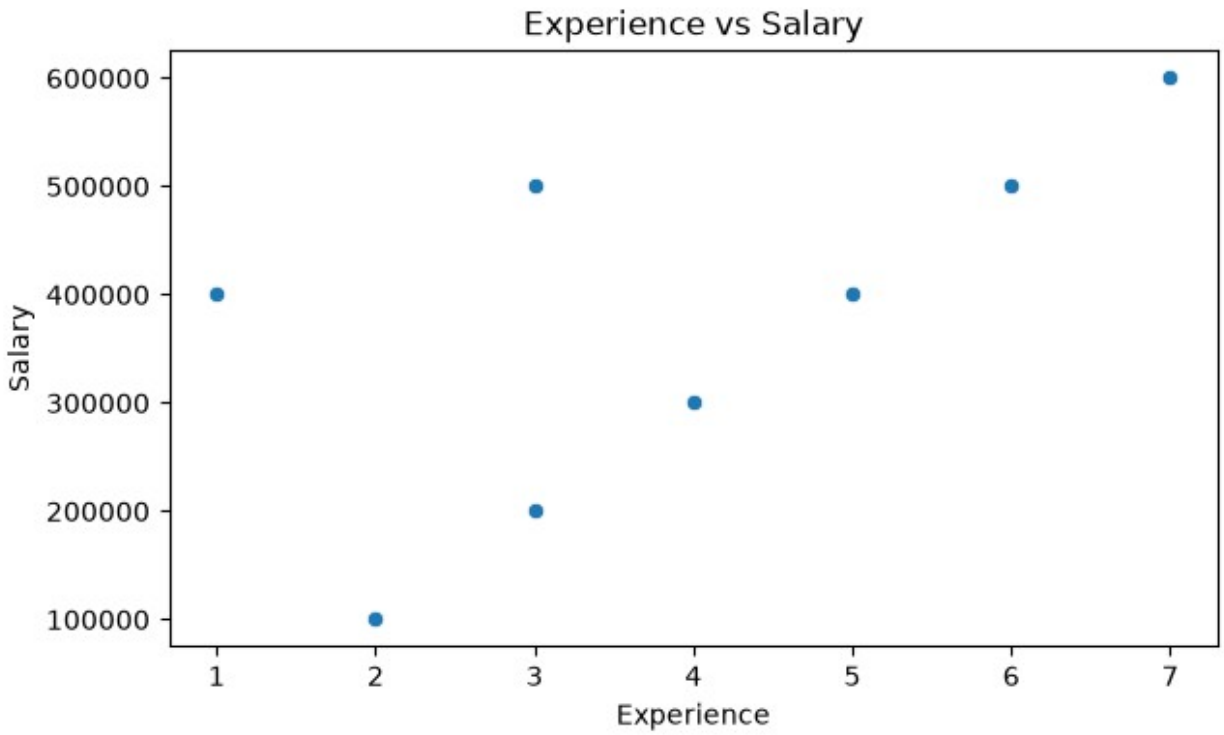
```
dtype: int64

Duplicate Rows: 0

Department Count:
Department
CSE          1
IT           1
CSM          1
DS           1
CHE          1
MECH         1
HR           1
SALES        1
MARKETING    1
Name: count, dtype: int64
```




Salary by Department
600000
500000
400000
300000
200000
100000
0
Salary
CSE
IT
CSM
DS
CHE
MECH
HR
SALES
MARKETING
Department


Experience vs Salary
600000
500000
400000
300000
200000
100000
Salary
1
2
3
4
5
6
7
Experience




```
Average Salary by Department:
Department
CHE          500000.0
CSE          100000.0
CSM          300000.0
DS           400000.0
HR           400000.0
IT           200000.0
MARKETING    500000.0
MECH         600000.0
SALES        100000.0
Name: Salary, dtype: float64
```

```python
#week-7

import pandas as pd
import matplotlib.pyplot as plt

# Load the employee dataset
df = pd.read_csv("employee.csv")

# Display the dataset
print("Employee Dataset:")
```

```
print(df.head())

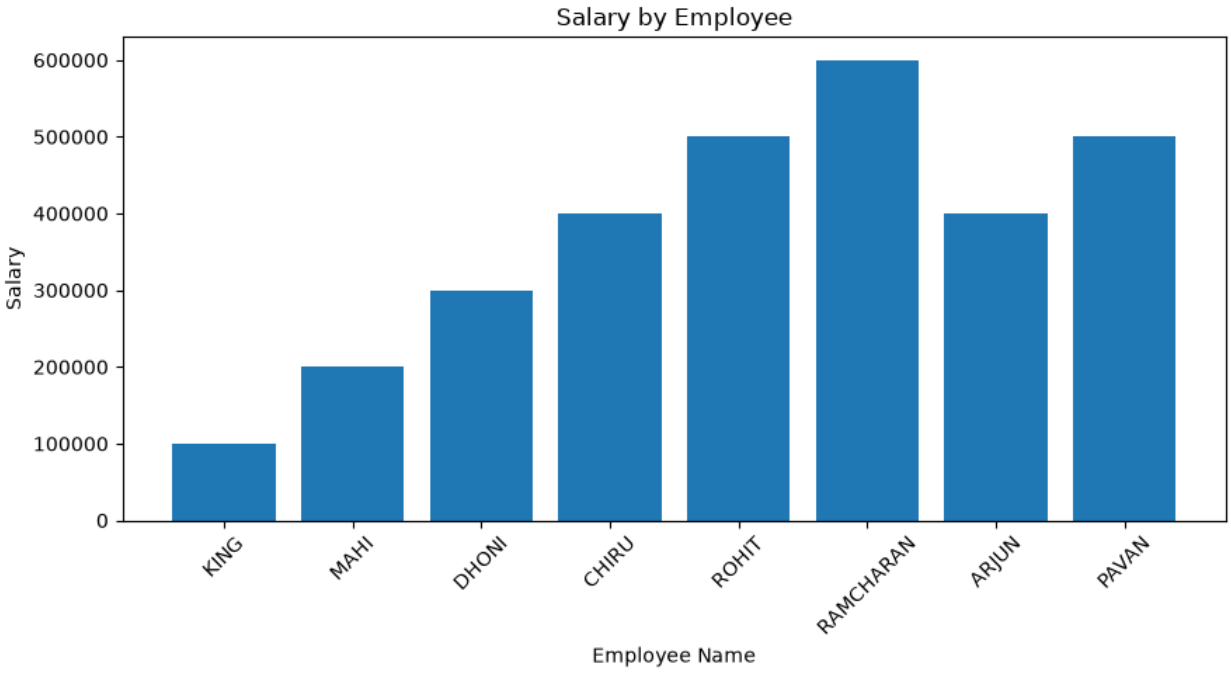
# 1. Line Plot: Employee ID vs Salary
plt.figure(figsize=(8, 5))
plt.plot(df['Emp_ID'], df['Salary'], marker='o')
plt.title("Employee ID vs Salary")
plt.xlabel("Employee ID")
plt.ylabel("Salary")
plt.grid(True)
plt.show()

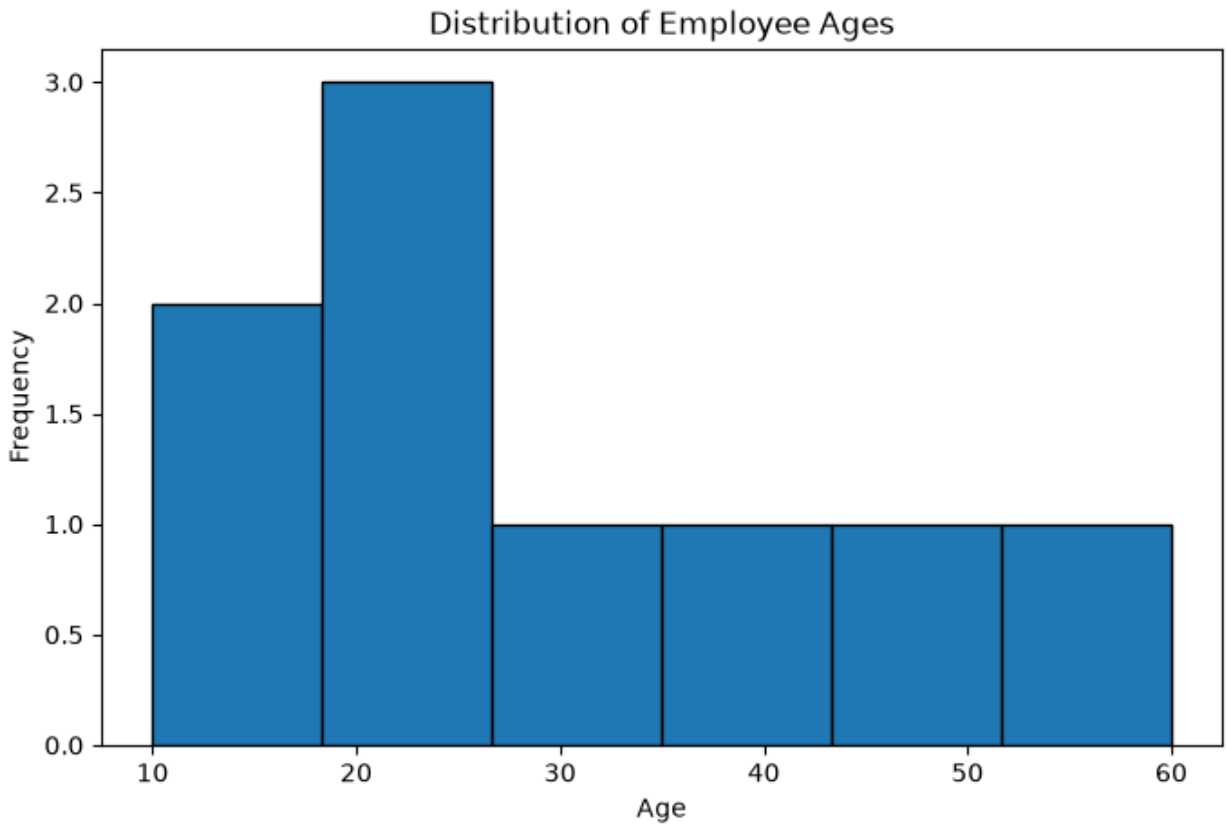
# 2. Bar Chart: Salary by Employee
plt.figure(figsize=(9, 5))
plt.bar(df['Name'], df['Salary'])
plt.title("Salary by Employee")
plt.xlabel("Employee Name")
plt.ylabel("Salary")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 3. Histogram: Employee Age Distribution
plt.figure(figsize=(8, 5))
plt.hist(df['Age'], bins=6, edgecolor='black')
plt.title("Distribution of Employee Ages")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```

```
Employee Dataset:
   Emp_ID   Name Department  Age  Salary  Experience
0       1   KING        CSE   10  100000           2
1       2   MAHI         IT   20  200000           3
2       3  DHONI        CSM   30  300000           4
3       4  CHIRU         DS   25  400000           5
4       5  ROHIT        CHE   40  500000           6
```


Employee ID vs Salary
600000
500000
400000
300000
200000
100000
Salary
1
2
3
4
5
6
7
8
9
Employee ID


Salary by Employee
600000
500000
400000
300000
200000
100000
0
Salary
KING
MAHI
DHONI
CHIRU
ROHIT
RAMCHARAN
ARJUN
PAVAN
Employee Name




```
#week - 8

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the employee dataset
df = pd.read_csv("employee.csv")

print("Employee Dataset:")
print(df.head())

# Select numerical columns
numeric_df = df.select_dtypes(include="number")

# 1. Correlation Heatmap
plt.figure(figsize=(8, 6))

correlation = numeric_df.corr()

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
```

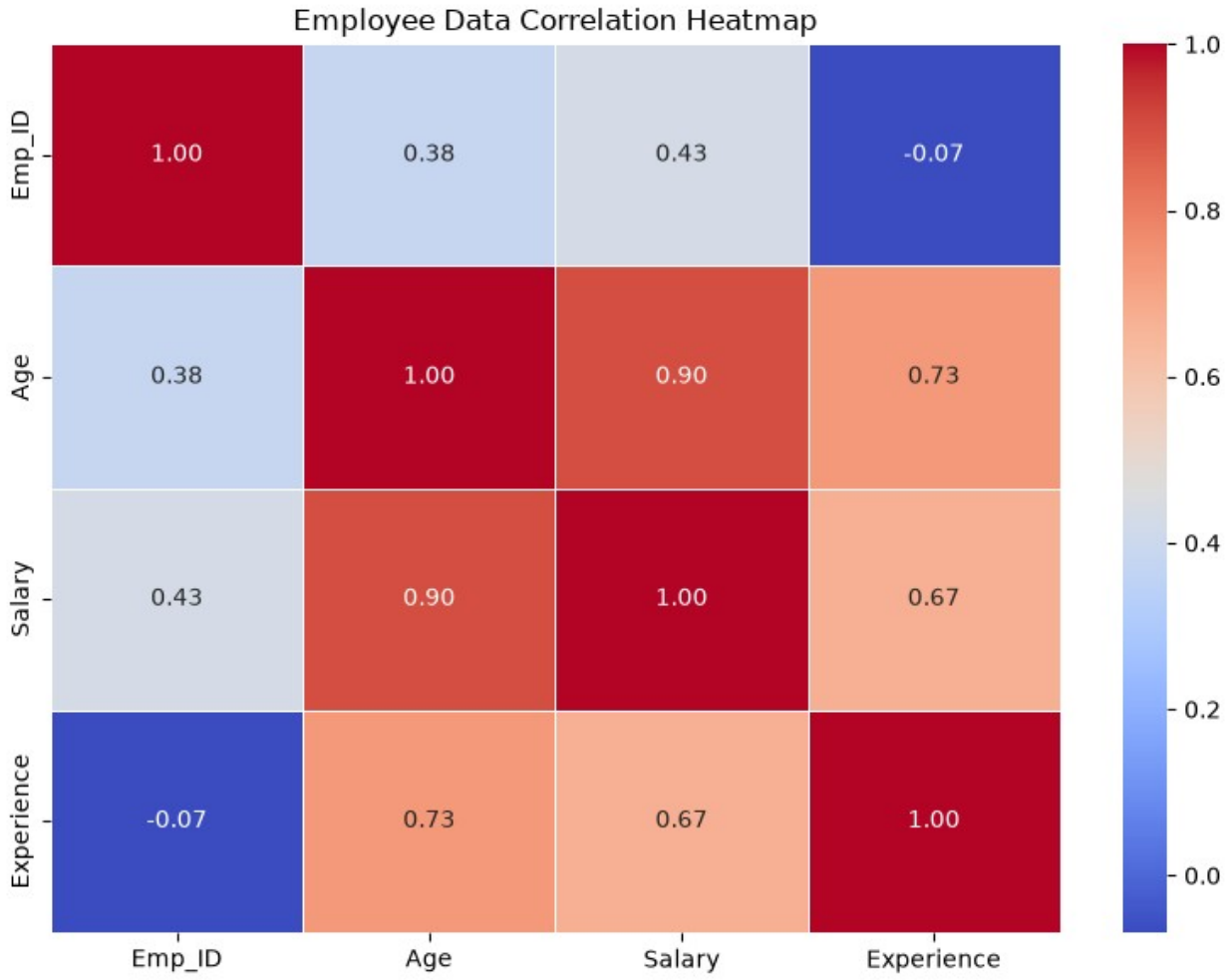
```
    fmt=".2f",
    linewidths=0.5
)

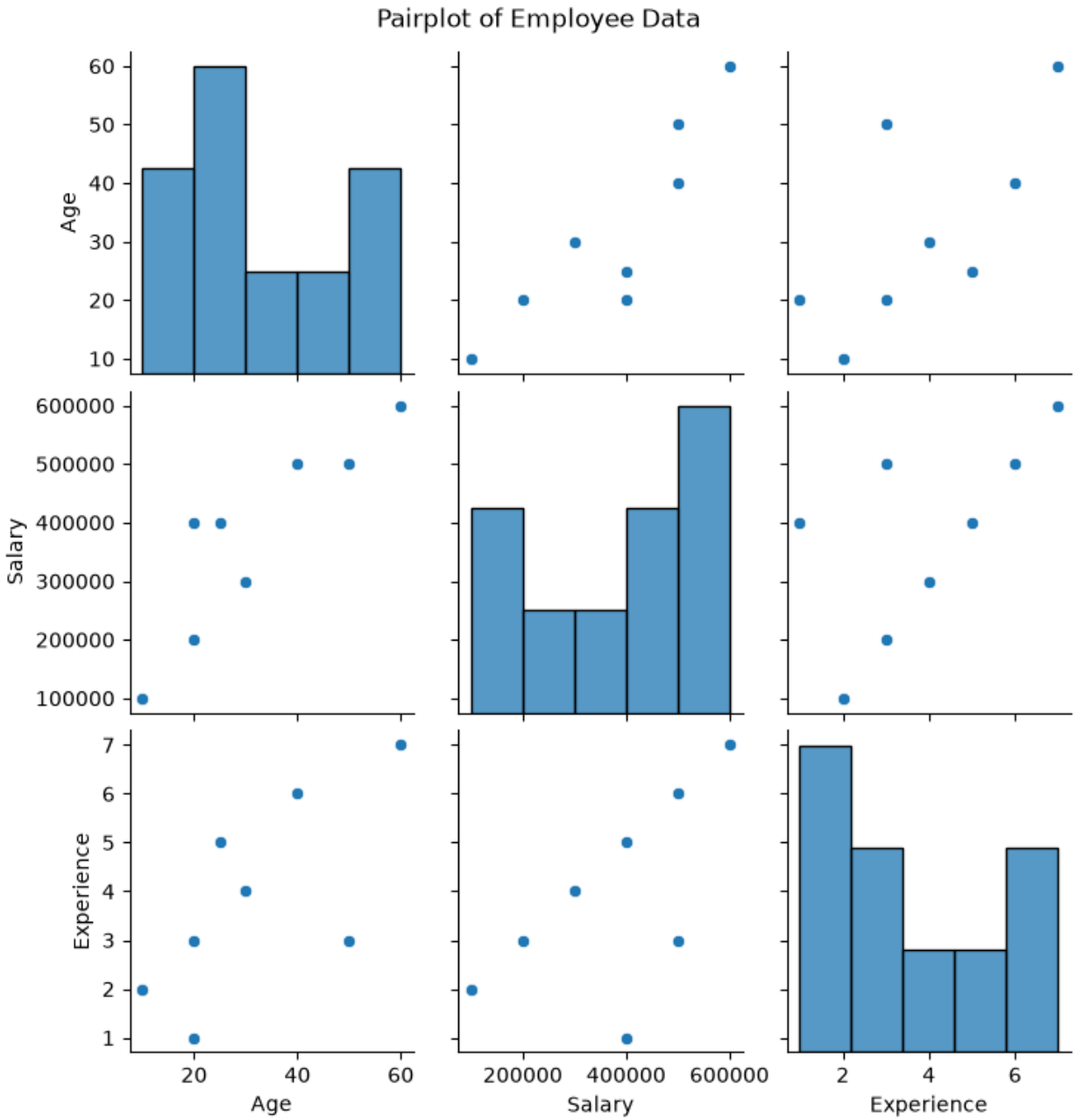
plt.title("Employee Data Correlation Heatmap")
plt.tight_layout()
plt.show()

# 2. Pairplot
sns.pairplot(
    df,
    vars=["Age", "Salary", "Experience"],
    diag_kind="hist"
)

plt.suptitle("Pairplot of Employee Data", y=1.02)
plt.show()
```

```
Employee Dataset:
   Emp_ID   Name Department  Age  Salary  Experience
0       1   KING        CSE   10  100000           2
1       2   MAHI         IT   20  200000           3
2       3  DHONI        CSM   30  300000           4
3       4  CHIRU         DS   25  400000           5
4       5  ROHIT        CHE   40  500000           6
```


Employee Data Correlation Heatmap
Emp_ID
Age
Salary
Experience
1.00
0.38
0.43
-0.07
0.38
1.00
0.90
0.73
0.43
0.90
1.00
0.67
-0.07
0.73
0.67
1.00
Emp_ID
Age
Salary
Experience
1.0
0.8
0.6
0.4
0.2
0.0




```
#week - 9

import pandas as pd
import plotly.express as px

# Load employee dataset
df = pd.read_csv("employee.csv")

print("Employee Dataset:")
print(df.head())
```

```python
# 1. Interactive Bar Chart: Salary by Employee
fig1 = px.bar(
    df,
    x="Name",
    y="Salary",
    color="Department",
    hover_data=["Emp_ID", "Age", "Experience"],
    title="Employee Salary Comparison",
    labels={"Salary": "Salary", "Name": "Employee Name"}
)
fig1.show()

# 2. Interactive Scatter Plot: Experience vs Salary
fig2 = px.scatter(
    df,
    x="Experience",
    y="Salary",
    color="Department",
    size="Age",
    hover_name="Name",
    hover_data=["Emp_ID"],
    title="Employee Experience vs Salary"
)
fig2.show()

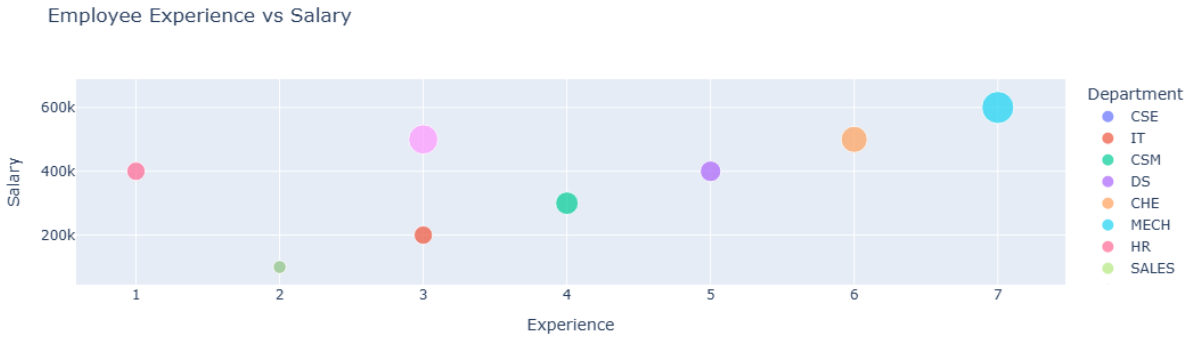
# 3. Interactive Histogram: Age Distribution
fig3 = px.histogram(
    df,
    x="Age",
    color="Department",
    nbins=10,
    title="Employee Age Distribution",
    labels={"Age": "Employee Age"}
)
fig3.show()

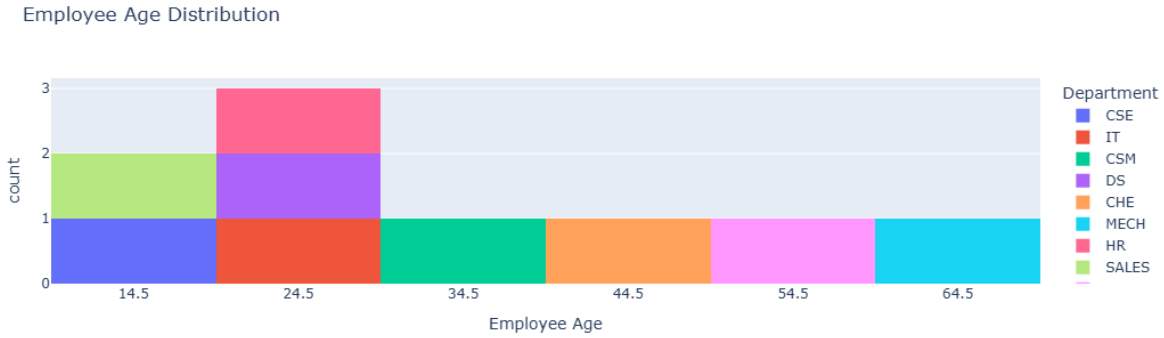
# 4. Interactive Box Plot: Salary Distribution
fig4 = px.box(
    df,
    x="Department",
    y="Salary",
    color="Department",
    hover_data=["Name", "Experience"],
    title="Salary Distribution by Department"
)
fig4.show()

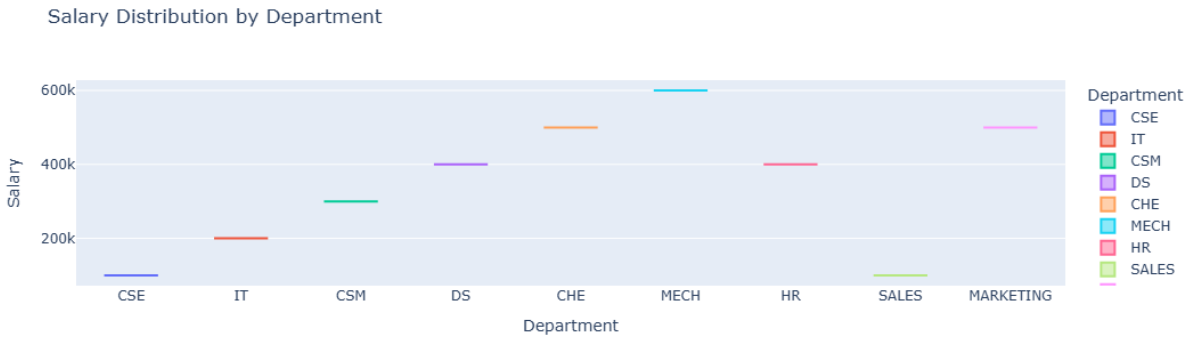
Employee Dataset:
   Emp_ID   Name Department  Age  Salary  Experience
0       1   KING        CSE   10  100000           2
```

1	2	MAHI	IT	20	200000	3
2	3	DHONI	CSM	30	300000	4
3	4	CHIRU	DS	25	400000	5
4	5	ROHIT	CHE	40	500000	6








Salary Distribution by Department
600k
400k
200k
Salary
CSE
IT
CSM
DS
CHE
MECH
HR
SALES
MARKETING
Department
Department
CSE
IT
CSM
DS
CHE
MECH
HR
SALES